Docker Claw (dclaw) -- Wire Protocol Specification v1

Field	Value
Version	1
Status	Draft
Date	2026-04-11
Transport	Unix Domain Sockets
Encoding	JSON-RPC 2.0 over UTF-8
Max Message	1 MB

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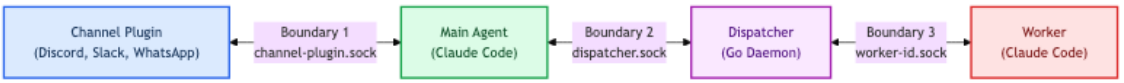
1. Overview

Docker Claw (dclaw) is a container-native multi-agent platform. Its architecture comprises four component types:

Component	Description	Lifecycle
Main Agent	Always-on container running Claude Code. Coordinates all work.	Long-lived
Worker Agent	Ephemeral container running Claude Code. Executes a scoped task.	Task-scoped
Channel Plugin	Independently versioned container. Bridges a messaging platform.	Long-lived
Dispatcher	Go binary daemon. Manages the worker fleet.	Long-lived

All inter-component communication travels over **Unix domain sockets** using **JSON-RPC 2.0** messages encoded as UTF-8.

There are three distinct communication boundaries, each with its own socket and message set:



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2. Transport Layer

Socket Paths

Each communication boundary uses a dedicated Unix domain socket. All sockets reside under `/var/run/dclaw/`.

Boundary	Socket Path Pattern	Creator
Channel Plugin <-> Main Agent	`/var/run/dclaw/channel-<plugin_name>.sock`	Main Agent
Main Agent <-> Dispatcher	`/var/run/dclaw/dispatcher.sock`	Dispatcher
Worker <-> Dispatcher	`/var/run/dclaw/worker-<worker_id>.sock`	Dispatcher

- The **Dispatcher** creates the main dispatcher socket at startup.
- The **Dispatcher** creates a per-worker socket at spawn time and bind-mounts it into the worker container.
- The **Main Agent** creates per-channel-plugin sockets when channel plugins register.

Socket Permissions

All sockets are created with mode `0660`. The owning user is `dclaw` and the owning group is `dclaw`. Containers run as the `dclaw` user to ensure access.

Connection Lifecycle

1. The **listener** creates the socket and calls `listen()`.
2. The **connector** calls `connect()`.
3. Both sides exchange a `dclaw.handshake` message (see Section 4).
4. Normal RPC traffic flows.
5. Either side may close the connection. The listener removes the socket file on shutdown.

Stream Framing

Each JSON-RPC message is sent as a single newline-delimited JSON object (one message per line). Messages are terminated by a single `\n` (0x0A) byte. Receivers split on `\n` and parse each line as a complete JSON-RPC message.

```
{"jsonrpc":"2.0","method":"channel.message_received","params":{...},"id":1}\n
{"jsonrpc":"2.0","result":{...},"id":1}\n
```

---

3. Envelope Format (JSON-RPC 2.0)

All messages conform to the JSON-RPC 2.0 specification. There are three message types:

3.1 Request

A request expects a response. It always contains an `id` field.

```
{
  "jsonrpc": "2.0",
  "method": "worker.spawn",
  "params": {
    "task": "Run the test suite for the auth module",
    "workspace": "/home/user/project"
  },
  "id": 1
}
```

3.2 Response (Success)

A successful response carries the `result` field and the matching `id`.

```
{
  "jsonrpc": "2.0",
  "result": {
    "worker_id": "w-a1b2c3d4",
    "container_id": "sha256:e5f6a7b8..."
  },
  "id": 1
}
```

3.3 Response (Error)

An error response carries the `error` field and the matching `id`.

```
{
  "jsonrpc": "2.0",
  "error": {
    "code": -32003,
    "message": "Quota exceeded",
    "data": {
      "metric": "concurrent_workers",
      "current": 10,
      "limit": 10
    }
  },
  "id": 1
}
```

3.4 Notification

A notification is a request with no `id`. The sender does not expect a response.

```
{
  "jsonrpc": "2.0",
  "method": "worker.status_changed",
```

```
  "params": {
    "worker_id": "w-a1b2c3d4",
    "old_status": "running",
    "new_status": "done",
    "output": "All 47 tests passed.",
    "error": null,
    "cost_usd": 0.12
  }
}
```

3.5 ID Generation

Request IDs are monotonically increasing integers scoped to each socket connection. Each side maintains its own counter starting at `1`. This avoids collisions because requests and responses always flow in opposite directions on the same connection.

---

4. Versioning and Handshake

Protocol Version

The protocol version is a single integer. This specification defines version **1**.

Handshake Requirement

The **first message** sent on every newly established socket connection MUST be a `dclaw.handshake` request. No other messages may be sent or processed until the handshake completes successfully.

`dclaw.handshake`

Direction: Sent by the connector (the side that called `connect()`).

Type: Request (expects a response).

Request

Parameter	Type	Required	Description
`protocol_version`	int	yes	The highest protocol version the sender supports.
`component_type`	string	yes	One of: `main`, `dispatcher`, `worker`, `channel`.
`component_version`	string	yes	Semantic version of the component (e.g., `1.2.0`).
`component_id`	string	yes	UUID identifying this component instance.

```
{
  "jsonrpc": "2.0",
  "method": "dclaw.handshake",
  "params": {
    "protocol_version": 1,
    "component_type": "channel",
    "component_version": "0.4.2",
    "component_id": "f47ac10b-58cc-4372-a567-0e02b2c3d479"
```

```
  },
  "id": 1
}
```

Response (Success)

Field	Type	Description
`accepted`	boolean	`true` if the handshake succeeded.
`negotiated_version`	int	The protocol version both sides will use (min of both).

```
{
  "jsonrpc": "2.0",
  "result": {
    "accepted": true,
    "negotiated_version": 1
  },
  "id": 1
}
```

Response (Rejected)

If the listener cannot support any version compatible with the connector, it returns an error:

```
{
  "jsonrpc": "2.0",
  "error": {
    "code": -32600,
    "message": "Unsupported protocol version",
    "data": {
      "requested": 2,
      "supported": [1]
    }
  },
  "id": 1
}
```

After a rejected handshake, the listener closes the connection.

Version Negotiation Rules

1. The connector sends the **highest** protocol version it supports.
2. The listener replies with the **minimum** of both sides' highest supported version.
3. If the negotiated version is below the connector's minimum supported version, the connector should close the connection.
4. All subsequent messages on the connection use the negotiated version's message set.

---

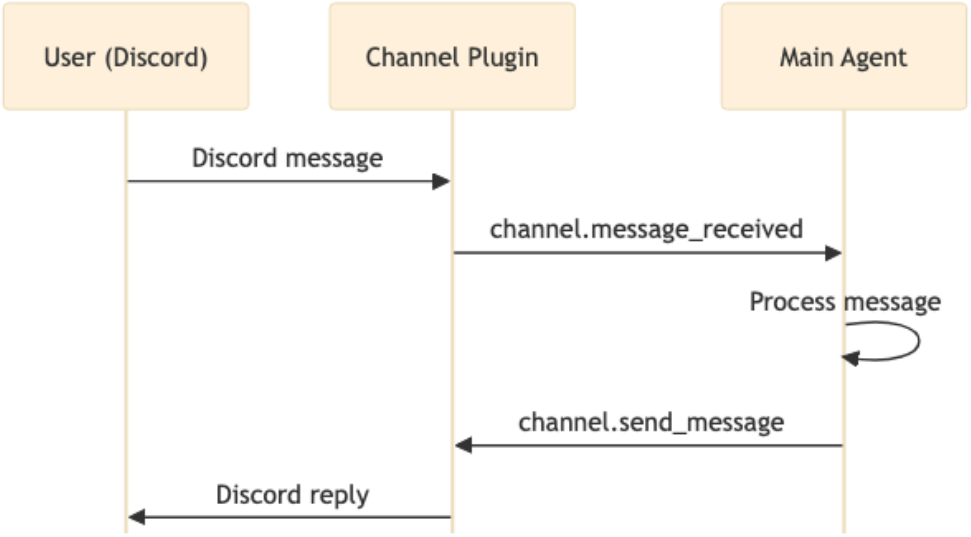
5. Boundary 1: Channel Plugin <-> Main Agent

Socket: `/var/run/dclaw/channel-<plugin_name>.sock`

This boundary normalizes platform-specific messaging APIs (Discord, Slack, WhatsApp, etc.) into a common format. The channel plugin translates between its native protocol and the dclaw wire protocol.

Channel Message Flow

The following diagram shows the typical message flow when a user sends a message through a channel plugin:



5.1 Channel Plugin -> Main Agent

These messages flow from the channel plugin to the main agent.

`channel.message_received`

Type: Notification (no response expected).

A user sent a message via the channel platform.

Parameter	Type	Required	Description
`channel_id`	string	yes	Platform-specific channel/conversation identifier.
`message_id`	string	yes	Platform-specific message identifier.
`user_id`	string	yes	Platform-specific user identifier.
`user_name`	string	yes	Display name of the user.
`text`	string	yes	Message body (plain text or markdown).
`attachments`	array	yes	List of attachments (may be empty). See below.
`timestamp`	string	yes	ISO 8601 timestamp of the message.
`channel_type`	string	yes	One of: `dm`, `group`.
`reply_to`	string	no	Message ID this message is replying to (for threads).

Attachment object:

Field	Type	Description
`name`	string	Filename of the attachment.
`type`	string	MIME type (e.g., `image/png`, `application/pdf`).
`size`	int	Size in bytes.
`url`	string	URL to download the attachment.

```
{
  "jsonrpc": "2.0",
  "method": "channel.message_received",
  "params": {
    "channel_id": "C-discord-981234567890",
    "message_id": "msg-1234567890",
    "user_id": "U-discord-112233445566",
    "user_name": "alice",
    "text": "Can you analyze NVDA for me?",
    "attachments": [],
    "timestamp": "2026-04-11T14:32:07Z",
    "channel_type": "dm",
    "reply_to": null
  }
}
```

Example with attachments:

```
{
  "jsonrpc": "2.0",
  "method": "channel.message_received",
  "params": {
    "channel_id": "C-slack-general",
    "message_id": "msg-9876543210",
    "user_id": "U-slack-W012A3CDE",
    "user_name": "bob",
    "text": "Here's the config file that's causing issues",
    "attachments": [
      {
        "name": "config.yaml",
        "type": "application/x-yaml",
        "size": 2048,
        "url": "https://files.slack.com/files-pri/T0123/config.yaml"
      }
    ],
    "timestamp": "2026-04-11T14:35:22Z",
    "channel_type": "group",
    "reply_to": "msg-9876543200"
```

```
  }
}
```

`channel.reaction_received`

Type: Notification (no response expected).

A user reacted to a message with an emoji.

Parameter	Type	Required	Description
`channel_id`	string	yes	Channel/conversation identifier.
`message_id`	string	yes	The message that was reacted to.
`user_id`	string	yes	The user who reacted.
`emoji`	string	yes	The emoji used (Unicode or shortcode).

```
{
  "jsonrpc": "2.0",
  "method": "channel.reaction_received",
  "params": {
    "channel_id": "C-discord-981234567890",
    "message_id": "msg-1234567890",
    "user_id": "U-discord-112233445566",
    "emoji": "thumbsup"
  }
}
```

`channel.status_changed`

Type: Notification (no response expected).

The channel plugin's connection status changed (connected, disconnected, or error).

Parameter	Type	Required	Description
`plugin_name`	string	yes	Name of the channel plugin (e.g., `discord`).
`version`	string	yes	Semantic version of the plugin.
`status`	string	yes	One of: `connected`, `disconnected`, `error`.
`error_message`	string	no	Human-readable error description (if status is `error`).

```
{
  "jsonrpc": "2.0",
  "method": "channel.status_changed",
  "params": {
    "plugin_name": "discord",
    "version": "0.4.2",
```

```
    "status": "connected",
    "error_message": null
  }
}
```

Example (error state):

```
{
  "jsonrpc": "2.0",
  "method": "channel.status_changed",
  "params": {
    "plugin_name": "slack",
    "version": "1.1.0",
    "status": "error",
    "error_message": "WebSocket connection lost: gateway timeout after 30s"
  }
}
```

5.2 Main Agent -> Channel Plugin

These messages flow from the main agent to the channel plugin. All are requests that expect a response.

`channel.send_message`

Type: Request.

Send a message to a channel.

Parameter	Type	Required	Description
`channel_id`	string	yes	Target channel/conversation.
`text`	string	yes	Message body.
`reply_to`	string	no	Message ID to reply to (for threading).
`files`	array	no	List of absolute file paths to attach.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "channel.send_message",
  "params": {
    "channel_id": "C-discord-981234567890",
    "text": "NVDA analysis complete. Current price: $142.50. Bullish setup
detected.",
    "reply_to": "msg-1234567890",
    "files": ["/tmp/dclaw/charts/nvda-daily.png"]
  },
```

```
  "id": 2
}
```

Response:

Field	Type	Description
`message_id`	string	The ID of the sent message.

```
{
  "jsonrpc": "2.0",
  "result": {
    "message_id": "msg-out-5555555555"
  },
  "id": 2
}
```

`channel.send_reaction`

Type: Request.

React to a message with an emoji.

Parameter	Type	Required	Description
`channel_id`	string	yes	Channel containing the message.
`message_id`	string	yes	Message to react to.
`emoji`	string	yes	Emoji to add.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "channel.send_reaction",
  "params": {
    "channel_id": "C-discord-981234567890",
    "message_id": "msg-1234567890",
    "emoji": "eyes"
  },
  "id": 3
}
```

Response:

```
{
  "jsonrpc": "2.0",
  "result": {},
```

```
  "id": 3
}
```

`channel.edit_message`

Type: Request.

Edit a previously sent message.

Parameter	Type	Required	Description
`channel_id`	string	yes	Channel containing the message.
`message_id`	string	yes	Message to edit.
`new_text`	string	yes	Replacement message body.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "channel.edit_message",
  "params": {
    "channel_id": "C-discord-981234567890",
    "message_id": "msg-out-5555555555",
    "new_text": "NVDA analysis complete. Current price: $142.50. Bullish setup
detected. Updated target: $155."
  },
  "id": 4
}
```

Response:

```
{
  "jsonrpc": "2.0",
  "result": {},
  "id": 4
}
```

`channel.fetch_history`

Type: Request.

Fetch recent messages from a channel.

Parameter	Type	Required	Description
`channel_id`	string	yes	Channel to fetch from.
`limit`	int	yes	Maximum number of messages to return.
`before`	string	no	Fetch messages before this message ID (pagination).

Request:

```
{
  "jsonrpc": "2.0",
  "method": "channel.fetch_history",
  "params": {
    "channel_id": "C-discord-981234567890",
    "limit": 20,
    "before": null
  },
  "id": 5
}
```

Response:

Field	Type	Description
`messages`	array	List of message objects (same shape as `channel.message_received` params).

```
{
  "jsonrpc": "2.0",
  "result": {
    "messages": [
      {
        "channel_id": "C-discord-981234567890",
        "message_id": "msg-1234567890",
        "user_id": "U-discord-112233445566",
        "user_name": "alice",
        "text": "Can you analyze NVDA for me?",
        "attachments": [],
        "timestamp": "2026-04-11T14:32:07Z",
        "channel_type": "dm",
        "reply_to": null
      },
      {
        "channel_id": "C-discord-981234567890",
        "message_id": "msg-out-5555555555",
        "user_id": "U-bot-dclaw",
        "user_name": "dclaw",
        "text": "NVDA analysis complete. Current price: $142.50. Bullish setup
detected.",
        "attachments": [],
        "timestamp": "2026-04-11T14:33:15Z",
        "channel_type": "dm",
        "reply_to": "msg-1234567890"
      }
    ]
  },
  "id": 5
}
```

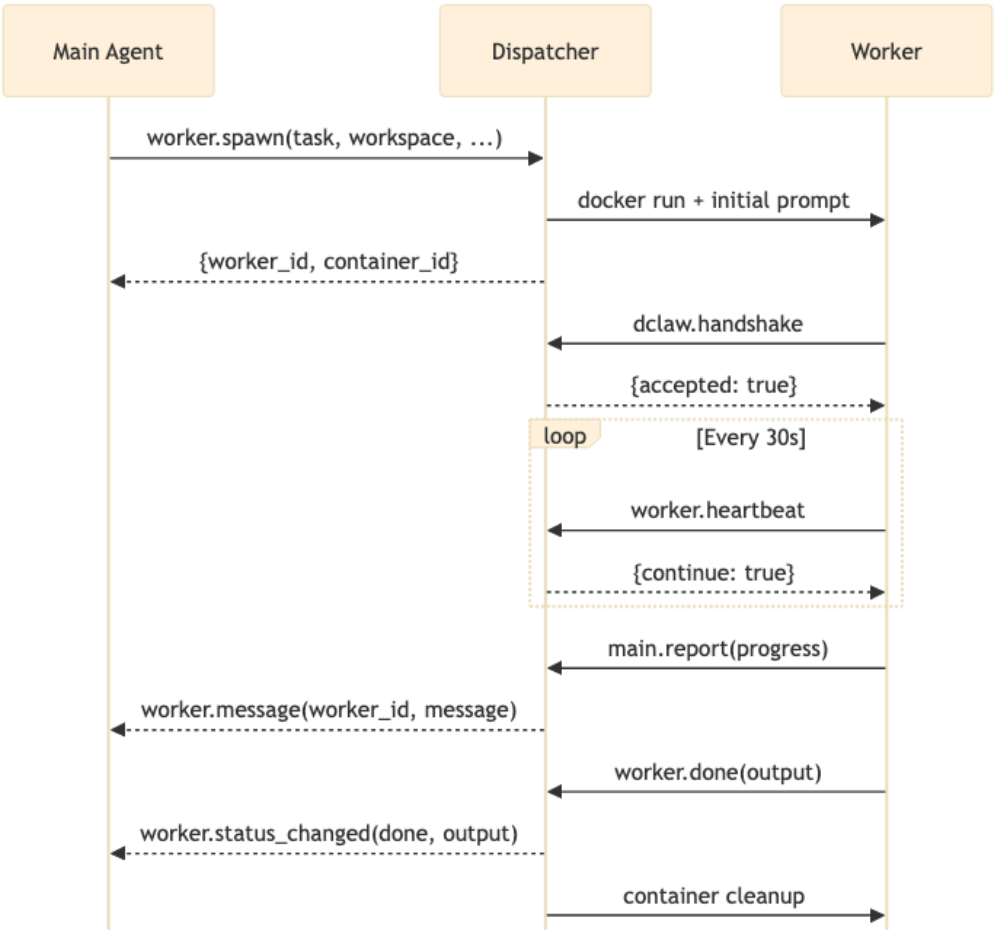
6. Boundary 2: Main Agent <-> Dispatcher

Socket: `/var/run/dclaw/dispatcher.sock`

This boundary allows the main agent to manage the worker fleet: spawning, querying, communicating with, and terminating worker containers.

Worker Spawn Lifecycle

The following diagram shows the full sequence for spawning a worker, from the initial request through heartbeats and reporting to completion:



6.1 Main Agent -> Dispatcher

These are RPC requests sent by the main agent to the dispatcher.

`worker.spawn`

Type: Request.

Create a new ephemeral worker container.

Parameter	Type	Required	Description

`task`	string	yes	The prompt/task description for the worker.
`workspace`	string	yes	Host path to bind-mount into the worker as its workspace.
`model`	string	no	Model to use (default: `opus`).
`tools`	array	no	List of allowed tool names. If omitted, all tools are allowed.
`egress_allowlist`	array	no	List of allowed hostnames for network egress. If omitted, no egress.
`timeout_seconds`	int	no	Maximum wall-clock time before auto-kill (default: `3600`).
`metadata`	object	no	Arbitrary key-value pairs for tracking/labeling.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "worker.spawn",
  "params": {
    "task": "Run the full test suite in /workspace/auth-service and report failures
with root cause analysis.",
    "workspace": "/home/user/projects/auth-service",
    "model": "opus",
    "tools": ["Bash", "Read", "Grep", "Glob"],
    "egress_allowlist": ["registry.npmjs.org", "github.com"],
    "timeout_seconds": 1800,
    "metadata": {
      "triggered_by": "discord:alice",
      "purpose": "test-run"
    }
  },
  "id": 10
}
```

Response:

Field	Type	Description
`worker_id`	string	Unique identifier for this worker.
`container_id`	string	Docker container ID.

```
{
  "jsonrpc": "2.0",
  "result": {
    "worker_id": "w-a1b2c3d4",
    "container_id": "sha256:e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4"
  },
  "id": 10
}
```

`worker.send_message`

Type: Request.

Send a message to a running worker (delivered via `worker.message_from_main` on Boundary 3).

Parameter	Type	Required	Description
`worker_id`	string	yes	Target worker.
`message`	string	yes	Message text to deliver.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "worker.send_message",
  "params": {
    "worker_id": "w-a1b2c3d4",
    "message": "Focus on the OAuth2 token refresh tests specifically."
  },
  "id": 11
}
```

Response:

Field	Type	Description
`ack`	boolean	`true` if the message was delivered.

```
{
  "jsonrpc": "2.0",
  "result": {
    "ack": true
  },
  "id": 11
}
```

`worker.get_status`

Type: Request.

Query a worker's current state.

Parameter	Type	Required	Description
`worker_id`	string	yes	Target worker.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "worker.get_status",
  "params": {
    "worker_id": "w-a1b2c3d4"
  },
  "id": 12
}
```

Response:

Field	Type	Description
`status`	string	One of: `queued`, `starting`, `running`, `done`, `failed`, `timeout`, `killed`.
`exit_code`	int	Exit code if `done` or `failed`. `null` otherwise.
`started_at`	string	ISO 8601 timestamp of when the worker started. `null` if queued.
`elapsed_seconds`	float	Wall-clock seconds since start. `null` if queued.
`cost_usd`	float	Estimated cost in USD so far.

```
{
  "jsonrpc": "2.0",
  "result": {
    "status": "running",
    "exit_code": null,
    "started_at": "2026-04-11T14:33:00Z",
    "elapsed_seconds": 127.4,
    "cost_usd": 0.08
  },
  "id": 12
}
```

`worker.list`

Type: Request.

List all workers, optionally filtered by status.

Parameter	Type	Required	Description
`status_filter`	string	no	Filter by status (e.g., `running`). If omitted, return all.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "worker.list",
  "params": {
```

```
    "status_filter": "running"
  },
  "id": 13
}
```

Response:

Field	Type	Description
`workers`	array	List of worker summary objects. See below.

Worker summary object:

Field	Type	Description
`id`	string	Worker ID.
`status`	string	Current status.
`task`	string	First 200 characters of the task.
`started_at`	string	ISO 8601 timestamp.
`cost_usd`	float	Estimated cost in USD so far.

```
{
  "jsonrpc": "2.0",
  "result": {
    "workers": [
      {
        "id": "w-a1b2c3d4",
        "status": "running",
        "task": "Run the full test suite in /workspace/auth-service and report
failures with root cause analysis.",
        "started_at": "2026-04-11T14:33:00Z",
        "cost_usd": 0.08
      },
      {
        "id": "w-e5f6g7h8",
        "status": "running",
        "task": "Refactor the database migration scripts to use the new ORM
schema.",
        "started_at": "2026-04-11T14:30:00Z",
        "cost_usd": 0.15
      }
    ]
  },
  "id": 13
}
```

`worker.kill`

Type: Request.

Force-terminate a worker.

Parameter	Type	Required	Description
`worker_id`	string	yes	Target worker.
`reason`	string	no	Human-readable reason for termination.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "worker.kill",
  "params": {
    "worker_id": "w-a1b2c3d4",
    "reason": "User requested cancellation"
  },
  "id": 14
}
```

Response:

Field	Type	Description
`killed`	boolean	`true` if the worker was terminated.

```
{
  "jsonrpc": "2.0",
  "result": {
    "killed": true
  },
  "id": 14
}
```

`worker.get_output`

Type: Request.

Get the final output from a completed worker.

Parameter	Type	Required	Description
`worker_id`	string	yes	Target worker.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "worker.get_output",
```

```
  "params": {
    "worker_id": "w-a1b2c3d4"
  },
  "id": 15
}
```

Response:

Field	Type	Description
`output`	string	The worker's final output text.
`exit_code`	int	Exit code (0 = success).
`duration_seconds`	float	Total wall-clock runtime.
`cost_usd`	float	Total estimated cost in USD.

```
{
  "jsonrpc": "2.0",
  "result": {
    "output": "Test suite complete. 47/47 tests passed. No failures detected.
Coverage: 94.2%.",
    "exit_code": 0,
    "duration_seconds": 312.7,
    "cost_usd": 0.23
  },
  "id": 15
}
```

6.2 Dispatcher -> Main Agent (Notifications)

These are asynchronous notifications pushed from the dispatcher to the main agent. They have no `id` and expect no response.

`worker.status_changed`

Type: Notification.

A worker's lifecycle status changed.

Parameter	Type	Required	Description
`worker_id`	string	yes	The affected worker.
`old_status`	string	yes	Previous status.
`new_status`	string	yes	New status.
`output`	string	no	Final output if `new_status` is `done`.
`error`	string	no	Error description if `new_status` is `failed`.

`cost_usd`	float	yes	Estimated cost at the time of the status change.

```
{
  "jsonrpc": "2.0",
  "method": "worker.status_changed",
  "params": {
    "worker_id": "w-a1b2c3d4",
    "old_status": "running",
    "new_status": "done",
    "output": "Test suite complete. 47/47 tests passed. No failures detected.
Coverage: 94.2%.",
    "error": null,
    "cost_usd": 0.23
  }
}
```

Example (failure):

```
{
  "jsonrpc": "2.0",
  "method": "worker.status_changed",
  "params": {
    "worker_id": "w-e5f6g7h8",
    "old_status": "running",
    "new_status": "failed",
    "output": null,
    "error": "OOM killed: container exceeded 4GB memory limit",
    "cost_usd": 0.41
  }
}
```

`worker.message`

Type: Notification.

A worker sent a message to the main agent via `main.report` on Boundary 3.

Parameter	Type	Required	Description
`worker_id`	string	yes	The worker that sent the message.
`message`	string	yes	The message content.
`timestamp`	string	yes	ISO 8601 timestamp.

```
{
  "jsonrpc": "2.0",
  "method": "worker.message",
  "params": {
    "worker_id": "w-a1b2c3d4",
```

```
    "message": "Progress: 32/47 tests complete. 2 failures so far in
token_refresh_test.go.",
    "timestamp": "2026-04-11T14:36:45Z"
  }
}
```

`quota.warning`

Type: Notification.

The system is approaching a quota limit.

Parameter	Type	Required	Description
`metric`	string	yes	One of: `concurrent_workers`, `session_cost`, `total_cost`.
`current`	float	yes	Current value of the metric.
`limit`	float	yes	Configured limit.
`percent`	float	yes	Percentage of the limit consumed (0-100).

```
{
  "jsonrpc": "2.0",
  "method": "quota.warning",
  "params": {
    "metric": "session_cost",
    "current": 4.50,
    "limit": 5.00,
    "percent": 90.0
  }
}
```

---

7. Boundary 3: Worker <-> Dispatcher

Socket: `/var/run/dclaw/worker-<worker_id>.sock`

This boundary provides the worker with a minimal set of RPCs. Workers are scoped executors -- they can report results, ask questions, send heartbeats, and declare completion.

7.1 Worker -> Dispatcher

`main.report`

Type: Request.

Send a message or result to the main agent. This is the underlying RPC behind the `report_to_main()` tool available to workers.

Parameter	Type	Required	Description
`message`	string	yes	The message content.

`type`	string	yes	One of: `progress`, `result`, `error`, `question`.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "main.report",
  "params": {
    "message": "Found 3 failing tests in token_refresh_test.go. Root cause: expired
mock certificates.",
    "type": "progress"
  },
  "id": 1
}
```

Response:

Field	Type	Description
`ack`	boolean	`true` if the message was forwarded.

```
{
  "jsonrpc": "2.0",
  "result": {
    "ack": true
  },
  "id": 1
}
```

`main.ask`

Type: Request (synchronous, blocking).

Ask the main agent a question and block until the main agent responds. The dispatcher mediates this by forwarding the question to the main agent and holding the RPC open until the answer arrives.

Parameter	Type	Required	Description
`question`	string	yes	The question text.
`timeout_seconds`	int	no	Max seconds to wait for an answer (default: 60).

Request:

```
{
  "jsonrpc": "2.0",
  "method": "main.ask",
  "params": {
    "question": "The test database credentials in .env.test appear invalid. Should I
regenerate them or skip database tests?",
```

```
    "timeout_seconds": 120
  },
  "id": 2
}
```

Response:

Field	Type	Description
`answer`	string	The main agent's response text.

```
{
  "jsonrpc": "2.0",
  "result": {
    "answer": "Regenerate the test credentials. The expected format is in docs/test-
setup.md."
  },
  "id": 2
}
```

Error (timeout):

```
{
  "jsonrpc": "2.0",
  "error": {
    "code": -32005,
    "message": "Timeout waiting for main agent response",
    "data": {
      "timeout_seconds": 120
    }
  },
  "id": 2
}
```

`worker.heartbeat`

Type: Request.

Periodic health signal sent by the worker. The dispatcher tracks heartbeat timestamps to detect unresponsive workers.

Parameter	Type	Required	Description
`worker_id`	string	yes	This worker's ID.
`memory_mb`	int	yes	Current memory usage in megabytes.
`elapsed_seconds`	float	yes	Wall-clock seconds since task started.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "worker.heartbeat",
  "params": {
    "worker_id": "w-a1b2c3d4",
    "memory_mb": 512,
    "elapsed_seconds": 180.5
  },
  "id": 3
}
```

Response:

Field	Type	Description
`continue`	boolean	`true` to continue. `false` means the worker has been killed and should exit immediately.

```
{
  "jsonrpc": "2.0",
  "result": {
    "continue": true
  },
  "id": 3
}
```

Response (kill signal via heartbeat):

```
{
  "jsonrpc": "2.0",
  "result": {
    "continue": false
  },
  "id": 3
}
```

`worker.done`

Type: Request.

The worker declares itself finished. After sending this message and receiving the acknowledgment, the worker should exit.

Parameter	Type	Required	Description
`output`	string	yes	Final result/output text.
`exit_code`	int	yes	Exit code. `0` = success, `1` = error.

Request:

```
{
  "jsonrpc": "2.0",
  "method": "worker.done",
  "params": {
    "output": "Test suite complete. 47/47 tests passed after regenerating test
credentials. Coverage: 94.2%.",
    "exit_code": 0
  },
  "id": 4
}
```

Response:

```
{
  "jsonrpc": "2.0",
  "result": {},
  "id": 4
}
```

7.2 Dispatcher -> Worker (Notifications)

These are asynchronous notifications pushed from the dispatcher to the worker. They have no `id` and expect no response.

`worker.message_from_main`

Type: Notification.

The main agent sent a message to this worker (originated from `worker.send_message` on Boundary 2).

Parameter	Type	Required	Description
`message`	string	yes	The message from the main agent.
`timestamp`	string	yes	ISO 8601 timestamp.

```
{
  "jsonrpc": "2.0",
  "method": "worker.message_from_main",
  "params": {
    "message": "Focus on the OAuth2 token refresh tests specifically.",
    "timestamp": "2026-04-11T14:35:00Z"
  }
}
```

`worker.kill_signal`

Type: Notification.

The dispatcher is terminating this worker. The worker should perform any minimal cleanup and exit promptly.

Parameter	Type	Required	Description
`reason`	string	yes	One of: `timeout`, `user_killed`, `quota_exceeded`.

```
{
  "jsonrpc": "2.0",
  "method": "worker.kill_signal",
  "params": {
    "reason": "timeout"
  }
}
```

---

8. Error Codes

8.1 Standard JSON-RPC 2.0 Error Codes

Code	Name	Description
-32700	Parse error	Invalid JSON received.
-32600	Invalid request	The JSON is not a valid JSON-RPC 2.0 request object.
-32601	Method not found	The requested method does not exist on this boundary.
-32602	Invalid params	Invalid or missing method parameters.
-32603	Internal error	Unexpected internal error in the receiver.

8.2 Custom dclaw Error Codes

Code	Name	Description
-32001	Worker not found	The specified `worker_id` does not exist.
-32002	Worker not running	The worker exists but is no longer running (status: `done`, `failed`, `timeout`, or `killed`).
-32003	Quota exceeded	A quota limit has been reached. No new workers can be spawned or the operation is rejected.
-32004	Spawn failed	Docker failed to create the container (image pull error, resource limits, etc.).
-32005	Timeout	The operation timed out (e.g., `main.ask` waiting for main agent response).
-32006	Channel plugin not connected	The target channel plugin is not currently connected.

8.3 Error Response Format

All error responses follow the JSON-RPC 2.0 error object structure. The optional `data` field provides structured context.

```
{
  "jsonrpc": "2.0",
  "error": {
    "code": -32001,
    "message": "Worker not found",
    "data": {
      "worker_id": "w-nonexistent"
    }
  },
  "id": 99
}
```

```
{
  "jsonrpc": "2.0",
  "error": {
    "code": -32004,
    "message": "Spawn failed",
    "data": {
      "docker_error": "image pull failed: registry.example.com/dclaw-worker:latest:
unauthorized",
      "exit_code": 125
    }
  },
  "id": 10
}
```

```
{
  "jsonrpc": "2.0",
  "error": {
    "code": -32006,
    "message": "Channel plugin not connected",
    "data": {
      "plugin_name": "slack",
      "last_seen": "2026-04-11T13:55:00Z"
    }
  },
  "id": 2
}
```

---

9. Heartbeat and Liveness

9.1 Worker Heartbeat

Workers send a `worker.heartbeat` request every **30 seconds** on Boundary 3.

9.2 Missed Heartbeat Detection

The dispatcher tracks the timestamp of each worker's last heartbeat. If **3 consecutive heartbeats are missed** (90 seconds with no heartbeat), the dispatcher:

1. Marks the worker's status as `failed`.
2. Sends a `worker.status_changed` notification to the main agent with:
   - `new_status`: `failed`
   - `error`: `"Heartbeat timeout: no heartbeat received for 90 seconds"`
3. Terminates the worker container (`docker kill`).
4. Cleans up the worker socket file.

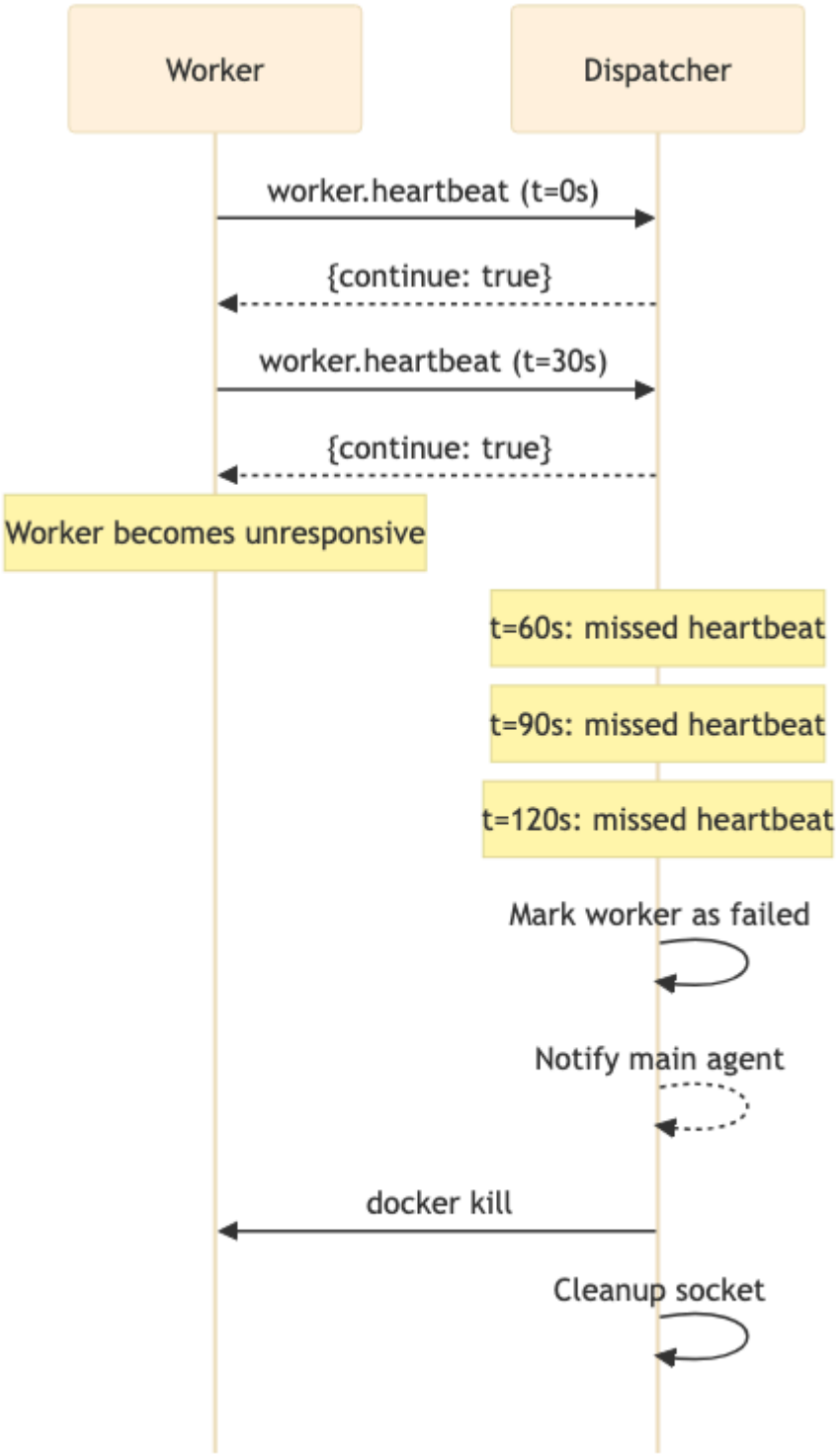
9.3 Heartbeat Response as Kill Signal

The dispatcher can use the heartbeat response to signal termination. When `continue` is `false` in the heartbeat response, the worker must:

1. Stop any in-progress work.
2. Optionally send a final `worker.done` with `exit_code: 1`.
3. Exit the process.

If the worker does not exit within **10 seconds** of receiving `continue: false`, the dispatcher force-kills the container.

9.4 Heartbeat Timeline



---

10. Message Size Limits

10.1 Maximum Message Size

Each JSON-RPC message must not exceed **1 MB** (1,048,576 bytes) when serialized to UTF-8.

10.2 File Content

File content MUST NOT be sent inline within JSON-RPC messages. Instead, pass files by **absolute path reference**:

- In `channel.send_message`, the `files` parameter contains absolute paths.
- Workers read and write files to their bind-mounted workspace directory.
- The `output` field in `worker.done` and `worker.get_output` should contain a text summary, not raw file contents. If detailed output is needed, write it to a file in the workspace and reference the path.

10.3 Oversized Message Handling

If a receiver receives a message exceeding 1 MB, it MUST:

1. Discard the message.
2. If the message was a request (had an `id`), respond with:

```
{
  "jsonrpc": "2.0",
  "error": {
    "code": -32600,
    "message": "Message exceeds 1MB size limit",
    "data": {
      "size_bytes": 1523741,
      "limit_bytes": 1048576
    }
  },
  "id": 42
}
```

3. If the message was a notification (no `id`), silently discard it and log a warning.

---

11. Concurrency Model

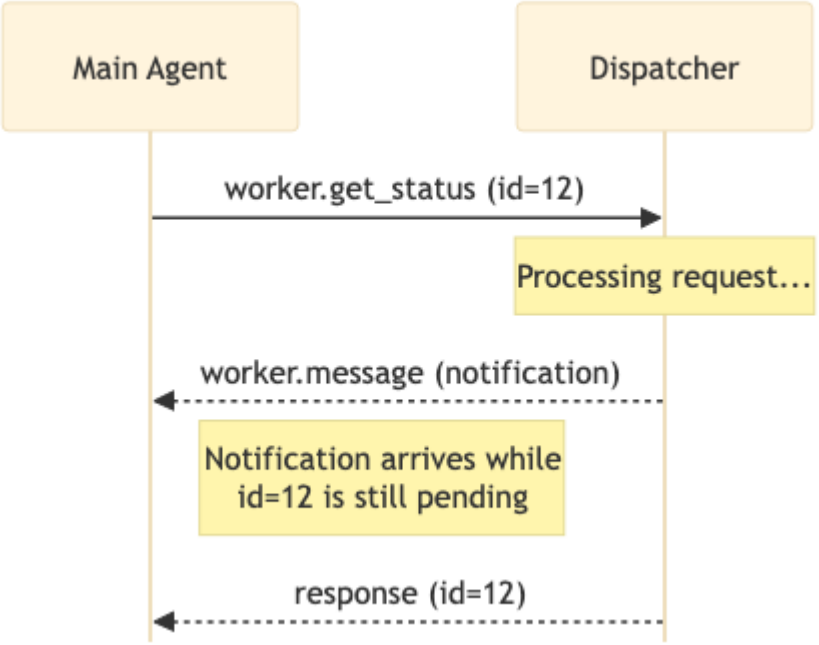
11.1 Request-Response Ordering (v1)

In protocol version 1, each socket connection handles **one request at a time**. The sender MUST wait for a response before sending the next request. This eliminates the need for request multiplexing or out-of-order response handling.

11.2 Notifications Are Exempt

Notifications (messages without an `id`) can arrive **at any time**, including while the receiver is waiting for a response to a pending request. Receivers must be prepared to handle interleaved notifications.

11.3 Example: Interleaved Notification



11.4 Connection Pooling

Components SHOULD maintain a single socket connection per boundary. If the connection drops, the connector should attempt reconnection with exponential backoff:

Attempt	Delay
1	100 ms
2	200 ms
3	400 ms
4	800 ms
5	1600 ms
6+	5000 ms

After 30 consecutive failed reconnection attempts, the component should log a fatal error and exit.

11.5 Future Versions

Protocol version 2 may introduce request multiplexing (multiple in-flight requests on a single connection). This will be negotiated at handshake time via the `negotiated_version` field.

---

12. Quick Reference

12.1 All Message Types

Method	Boundary	Direction	Type

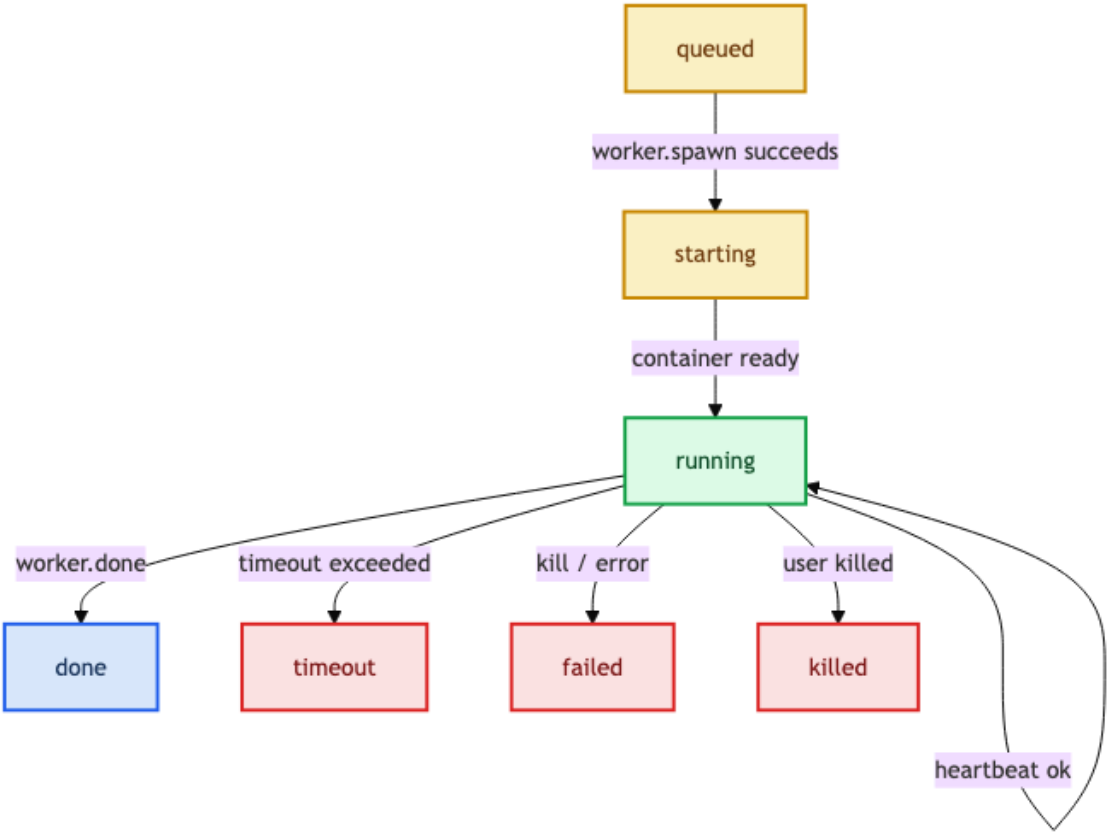
`dclaw.handshake`	All	Connector -> Listener	Request
`channel.message_received`	Channel Plugin <-> Main Agent	Plugin -> Main	Notification
`channel.reaction_received`	Channel Plugin <-> Main Agent	Plugin -> Main	Notification
`channel.status_changed`	Channel Plugin <-> Main Agent	Plugin -> Main	Notification
`channel.send_message`	Channel Plugin <-> Main Agent	Main -> Plugin	Request
`channel.send_reaction`	Channel Plugin <-> Main Agent	Main -> Plugin	Request
`channel.edit_message`	Channel Plugin <-> Main Agent	Main -> Plugin	Request
`channel.fetch_history`	Channel Plugin <-> Main Agent	Main -> Plugin	Request
`worker.spawn`	Main Agent <-> Dispatcher	Main -> Dispatcher	Request
`worker.send_message`	Main Agent <-> Dispatcher	Main -> Dispatcher	Request
`worker.get_status`	Main Agent <-> Dispatcher	Main -> Dispatcher	Request
`worker.list`	Main Agent <-> Dispatcher	Main -> Dispatcher	Request
`worker.kill`	Main Agent <-> Dispatcher	Main -> Dispatcher	Request
`worker.get_output`	Main Agent <-> Dispatcher	Main -> Dispatcher	Request
`worker.status_changed`	Main Agent <-> Dispatcher	Dispatcher -> Main	Notification
`worker.message`	Main Agent <-> Dispatcher	Dispatcher -> Main	Notification
`quota.warning`	Main Agent <-> Dispatcher	Dispatcher -> Main	Notification
`main.report`	Worker <-> Dispatcher	Worker -> Dispatcher	Request
`main.ask`	Worker <-> Dispatcher	Worker -> Dispatcher	Request
`worker.heartbeat`	Worker <-> Dispatcher	Worker -> Dispatcher	Request
`worker.done`	Worker <-> Dispatcher	Worker -> Dispatcher	Request
`worker.message_from_main`	Worker <-> Dispatcher	Dispatcher -> Worker	Notification
`worker.kill_signal`	Worker <-> Dispatcher	Dispatcher -> Worker	Notification

12.2 Message Counts by Boundary

Boundary	Requests	Notifications	Total
Channel Plugin <-> Main Agent	4	3	7
Main Agent <-> Dispatcher	6	3	9
Worker <-> Dispatcher	4	2	6
Handshake (all boundaries)	1	0	1

Total	**15**	**8**	**23**

12.3 Worker Lifecycle State Machine



12.4 Socket Path Summary

Socket Path	Listener	Connector
`/var/run/dclaw/channel-<plugin_name>.sock`	Main Agent	Channel Plugin
`/var/run/dclaw/dispatcher.sock`	Dispatcher	Main Agent
`/var/run/dclaw/worker-<worker_id>.sock`	Dispatcher	Worker

12.5 Error Code Summary

Code	Name	Applicable Boundaries
-32700	Parse error	All
-32600	Invalid request	All
-32601	Method not found	All
-32602	Invalid params	All

-32603	Internal error	All
-32001	Worker not found	Boundary 2, Boundary 3
-32002	Worker not running	Boundary 2
-32003	Quota exceeded	Boundary 2
-32004	Spawn failed	Boundary 2
-32005	Timeout	Boundary 3
-32006	Channel plugin not connected	Boundary 1

End of specification.